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Swarm Intelligence Approaches for Parameter Setting of Deep Learning Neural Network: Case Study on Phishing Websites Classification

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Introduction

Introduction

With rapid growth of online services, we are witnessing a rapid increase of online security threats:

- Pharming
- Spoofing
- **Phishing**

Phishing attack happens when malicious website acts, looks and feels almost identical to the legitimate one.

Resulting in stolen:

- personal and financial information,
- login credentials and
- other sensitive data.

Introduction - detecting Phishing websites

In general, two most popular approaches are used for detecting Phishing websites:

- **Blacklist and whitelist based approaches**
  - usually cannot cover all phishing websites
  - takes a considerable amount of time before it is added to the list
- **Intelligent heuristic-based approaches**
  - can recognize freshly created phishing websites in real time

Introduction - Motivation and Goals

Many intelligent Phishing detection methods are suggested:

- self-structuring neural networks,
- complex hybrid models and
- different kind of neural networks.

In most cases neural networks fall behind the conventional classifiers (e.g. RF, DT, SVM).

- **The biggest drawback** is **finding** the right **learning parameters and topology**.

Main goal:

- study whether we could solve mentioned drawback by utilizing our proposed method based on swarm intelligence algorithms.

Proposed Method

Proposed Method

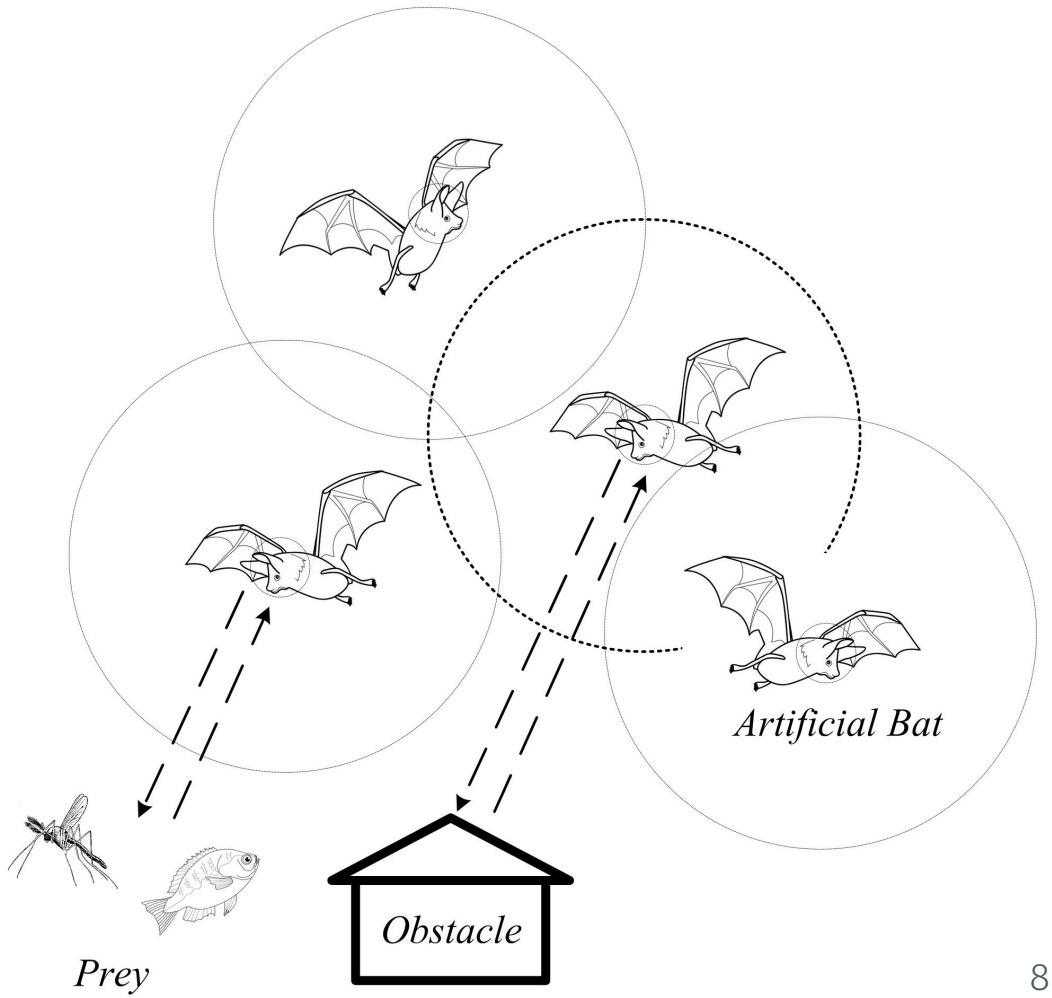
Proposed method, titled as TDLBA or TDLHBA (Tuning Deep Learning using Bat/Hybrid Bat Algorithm) combines:

- Swarm Intelligence Algorithms
- Tuning of Deep Learning parameters

Proposed Method

Swarm Intelligence Algorithms

- Bat Algorithm (BA)
- Hybrid Bat Algorithm (HBA)



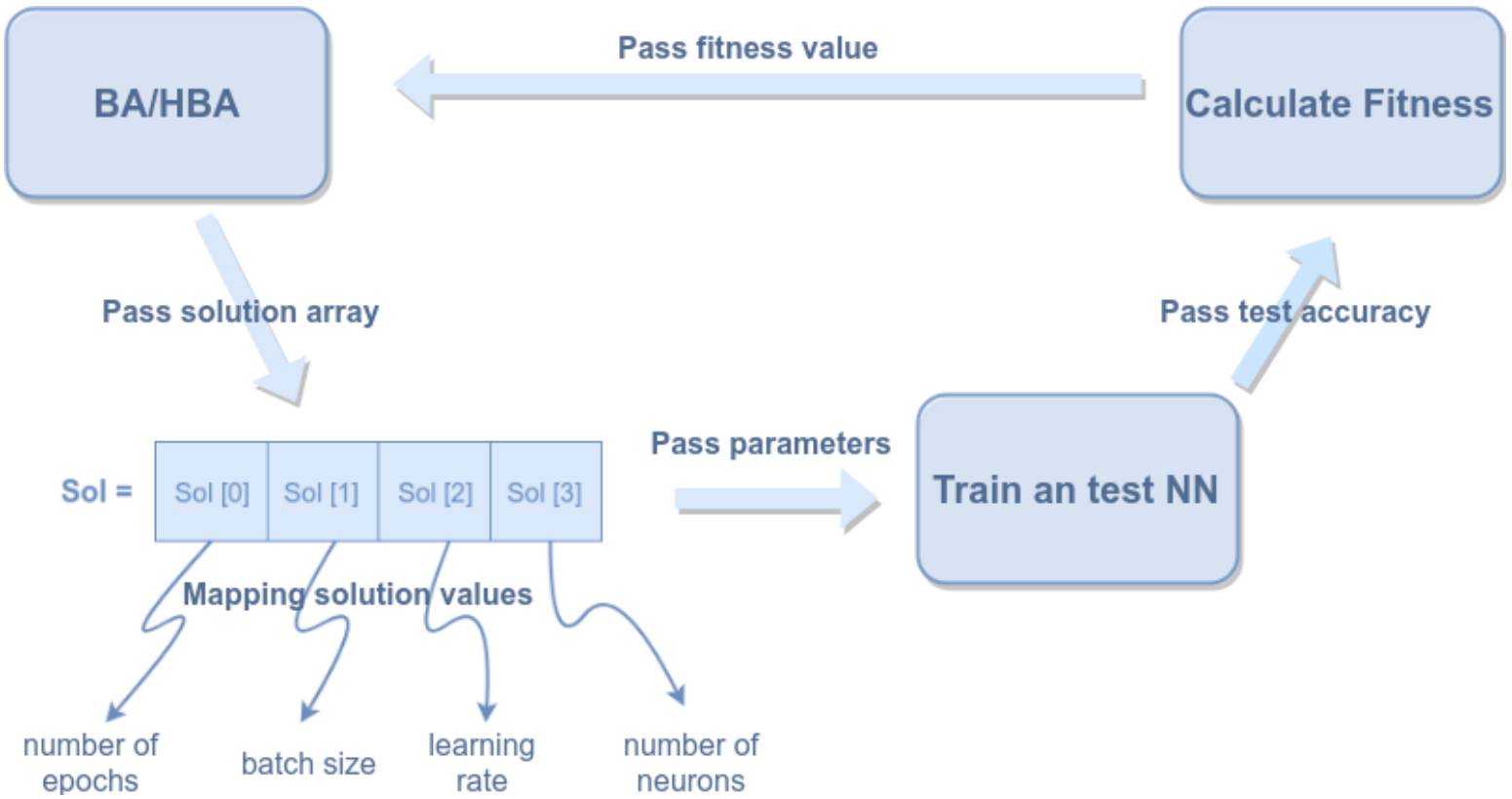
Proposed Method

Utilizing BA/HBA we're trying to find optimal value for the following parameters:

- number of epochs,
- batch size,
- learning rate and
- number of neurons in the first hidden layer.

Proposed Method

Conceptual diagram of our TDLBA/TDLHBA method:



Experiment

Dataset

Phishing Websites Data Set from the UCI Machine Learning Repository was used.

Dataset properties:

- Number of instances: 11,055
- Number of features: 31
- Number of classes: 2 classes
- Distribution of classes:
  - 3,793 phishing websites
  - 7,262 legitimate websites

Dataset - Features

All the features' values have been determined in accordance with a set of heuristic rules, devised to best reflect the presence of potentially phishy websites.

Some of the most important features:

- **Address bar based features:** Fake HTTPS and SSL, having @ symbol, using IP address, long URL ...
- **Abnormal based features:** Request URL, URL of anchor, server from handler, abnormal URL ...
- **HTML and JS based features:** Redirect page, disabling the right click, using pop-up window ...
- **Domain based features:** Age of domain, DNS record, web traffic ...

Experimental Settings - BA and HBA

BA and HBA were initialized using:

- Dimension of problem: 4 (number of parameters)
- Population size: 40
- Number of generations: 100
- Loudness: 0.5
- Pulse rate: 0.5
- Minimum frequency: 0.0
- Maximum frequency: 1.0
- Lower bound: 0.0
- Upper bound: 1.0

Additionally for HBA were set:

- F (Scaling factor): 0.5
- CR (Crossover probability): 0.5

We used a 20% validation split for calculating test accuracy needed for evaluation of fitness function.

Experimental Settings - Neural Network

Feed-forward NN with two fully connected hidden layers:

- Input layer's width is matching the the number of features (30).
- The width of first hidden layer is set dynamically.
- The number of neurons in second hidden layer is set to 30.
- Both hidden layers are using non-linear activation function ReLU.
- Output layer is using Sigmoid function.
- ADAM optimizer function is used.

Results

Results

We conducted two experiments:

- Assessment of the performance of the NN with handpicked training parameters.
- Utilizing our TDLBA/TDLHBA to obtain optimal parameter solution.

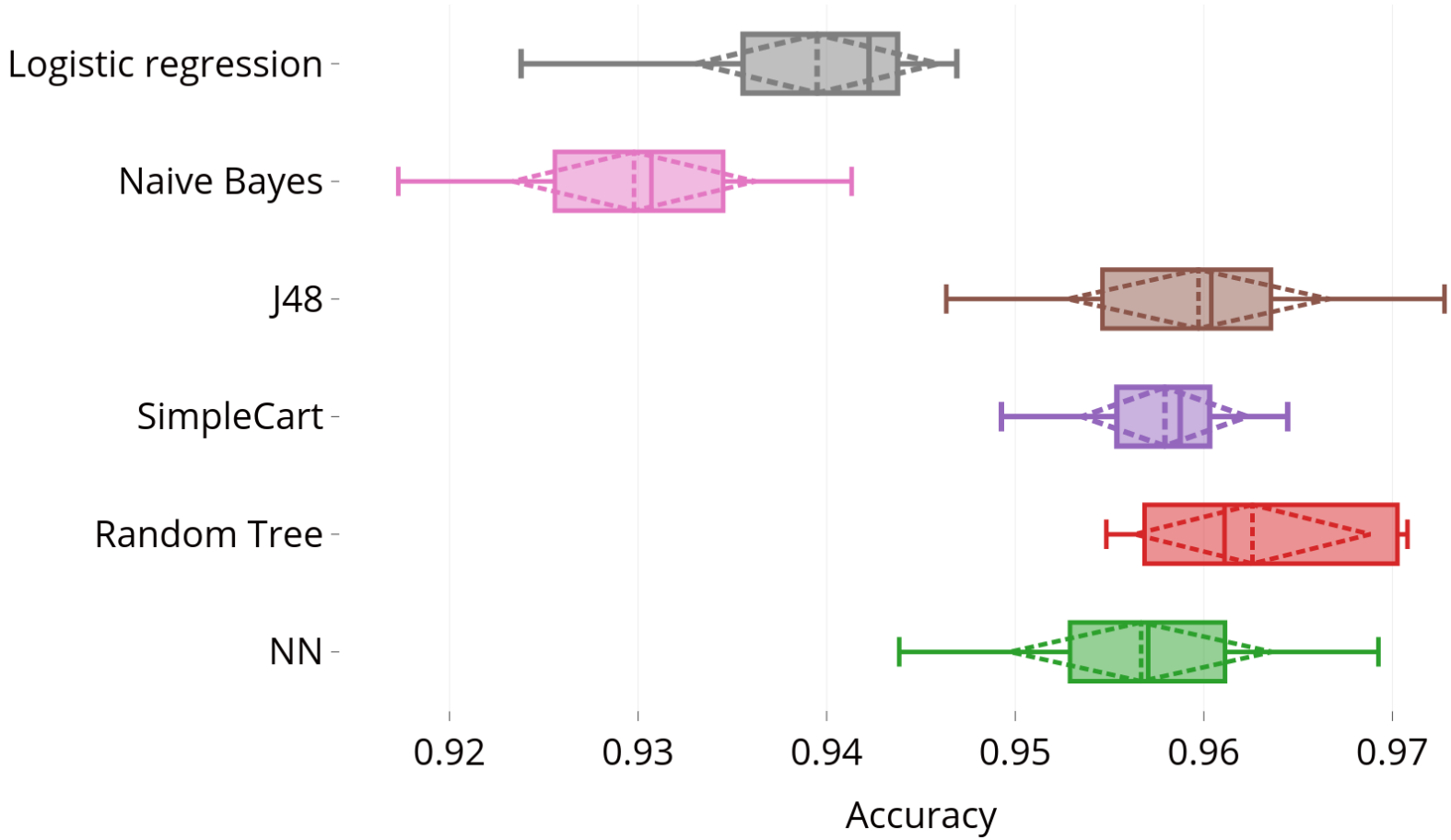
Validation was done using ten-fold cross validation procedure.

Neural Network Performance

Handpicked NN performance.

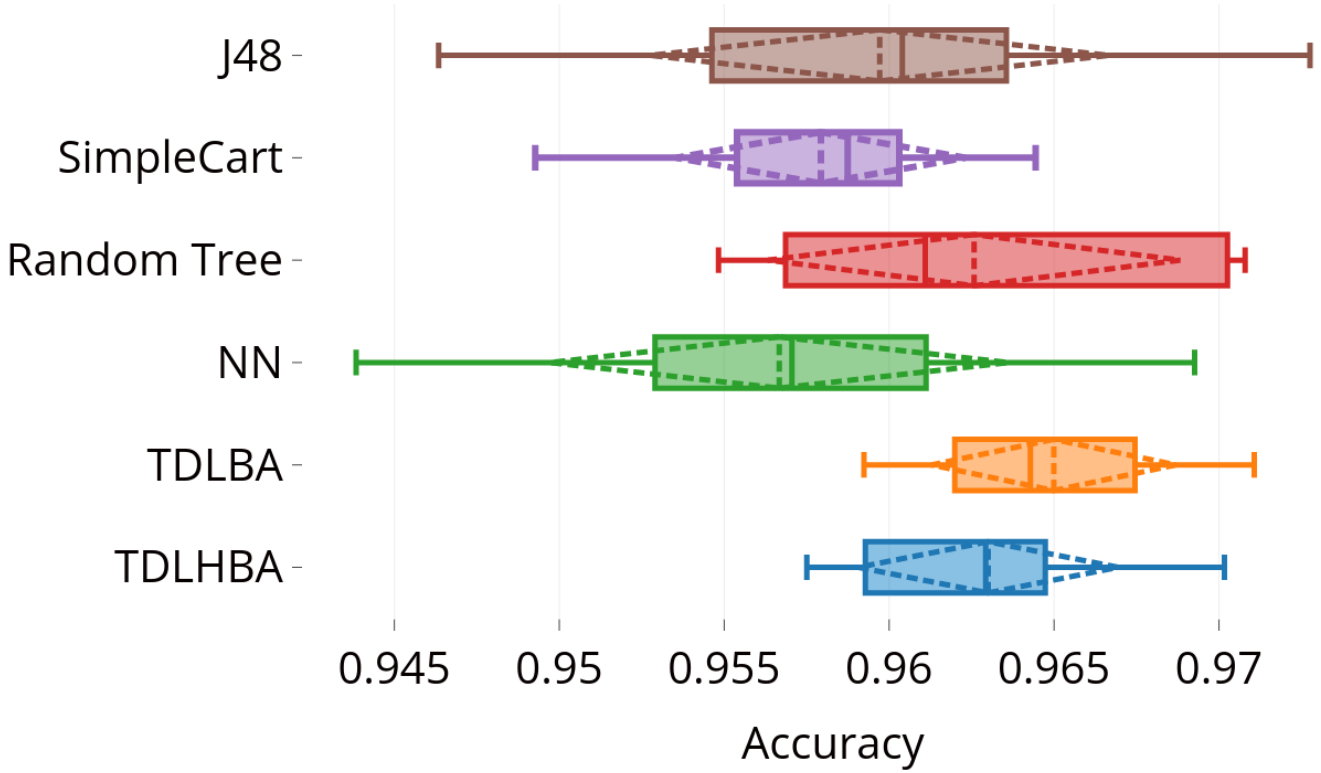
Parameters:

- Nr. of epochs: 150
- Batch size: 32
- Learning rate: 10^{-3}
- Number of neurons: 30



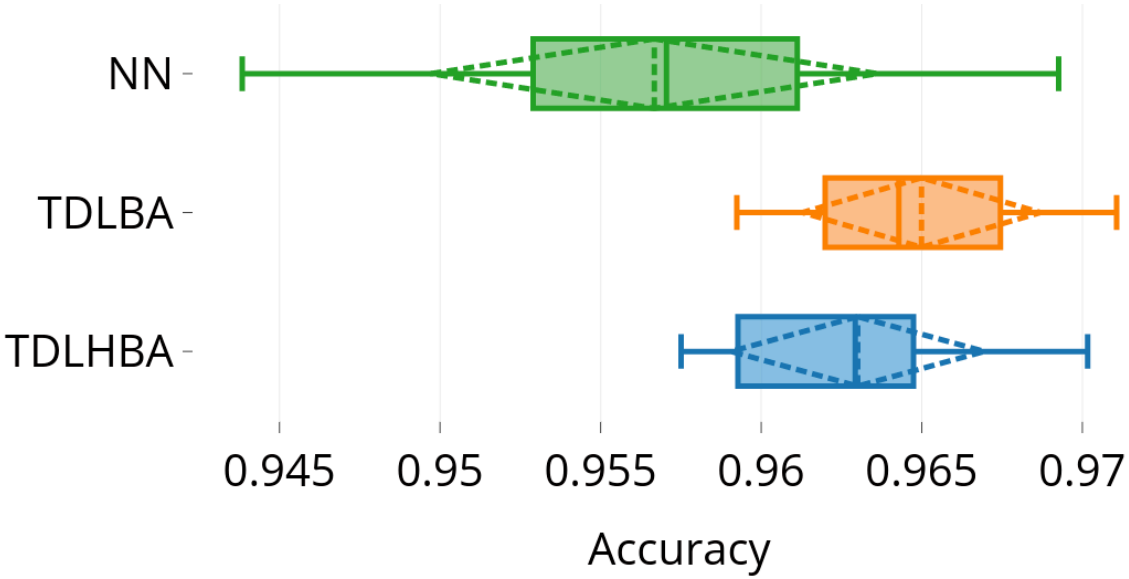
Improved Neural Network Performance

Performance comparison of NN's using parameter settings found utilizing TDLBA/TDLHBA.



Improved Neural Network Performance

Comparison of NN's performance:



Comparison of NN's parameters:

Parameter	NN	TDLBA	TDLHBA
Number of epoch	150	130	100
Batch size	32	52	10
Learning rate	10^{-3}	0.0043526	0.0017470
Number of neurons	30	35	40

Conclusion

Conclusion

We presented a novel method TDLBA/TDLHBA, utilizing a swarm intelligence based approach to set parameters of deep learning neural network.

- Promising results, outperforming traditional classifiers.
- The main advantage of the proposed method is very straightforward usage with various feed-forward NN topologies and different datasets.

Future Work

Based on encouraging results, we would like to expand our work:

- Utilizing different swarm intelligence algorithms.
- Performance benchmarking of proposed method on different datasets.
- Application of the main idea, the optimization of parameters to the other conventional classifiers.
- Self-construction of NN using swarm intelligence algorithms.

Thank you for your attention!